



AI & Data Science: Driving Business Transformation

The University of Tokyo Global Consumer Intelligence

June 2026



Introduction



Yuki Ishida

Director and Group CEO

Profile

- Mechanical and Aerospace Engineering, Cornell University
- Graduate School of Engineering, The University of Tokyo
- In 2006, joined Goldman Sachs at Asia Special Situations Group, overseeing Goldman's principal investment activities into private equity, corporate bond as well as real estate. Led the team's turnaround investment as the portfolio manager
- In 2011, founded YCP in Tokyo to provide management services for corporates in Japan. To grow as an Asia-focused global firm, YCP migrated its global headquarter to Hong Kong in 2014, and then onto Singapore in 2021.
- In 2021, YCP completed its IPO onto Tokyo Stock Exchange, as a leading Asia-focused consulting firm

Company Overview

- YCP started its business in August 2011 as Yamato Capital Partners (now YCP Japan)
- In 2014, to grow as an international enterprise, the firm moved its global headquarter to Hong Kong and was rebranded to YCP
- In 2021, YCP Holdings (Global) Limited was established in Singapore as the Group's parent company and made its IPO on the Growth Board of the Tokyo stock Exchange

15
Years of History

700+
Professionals

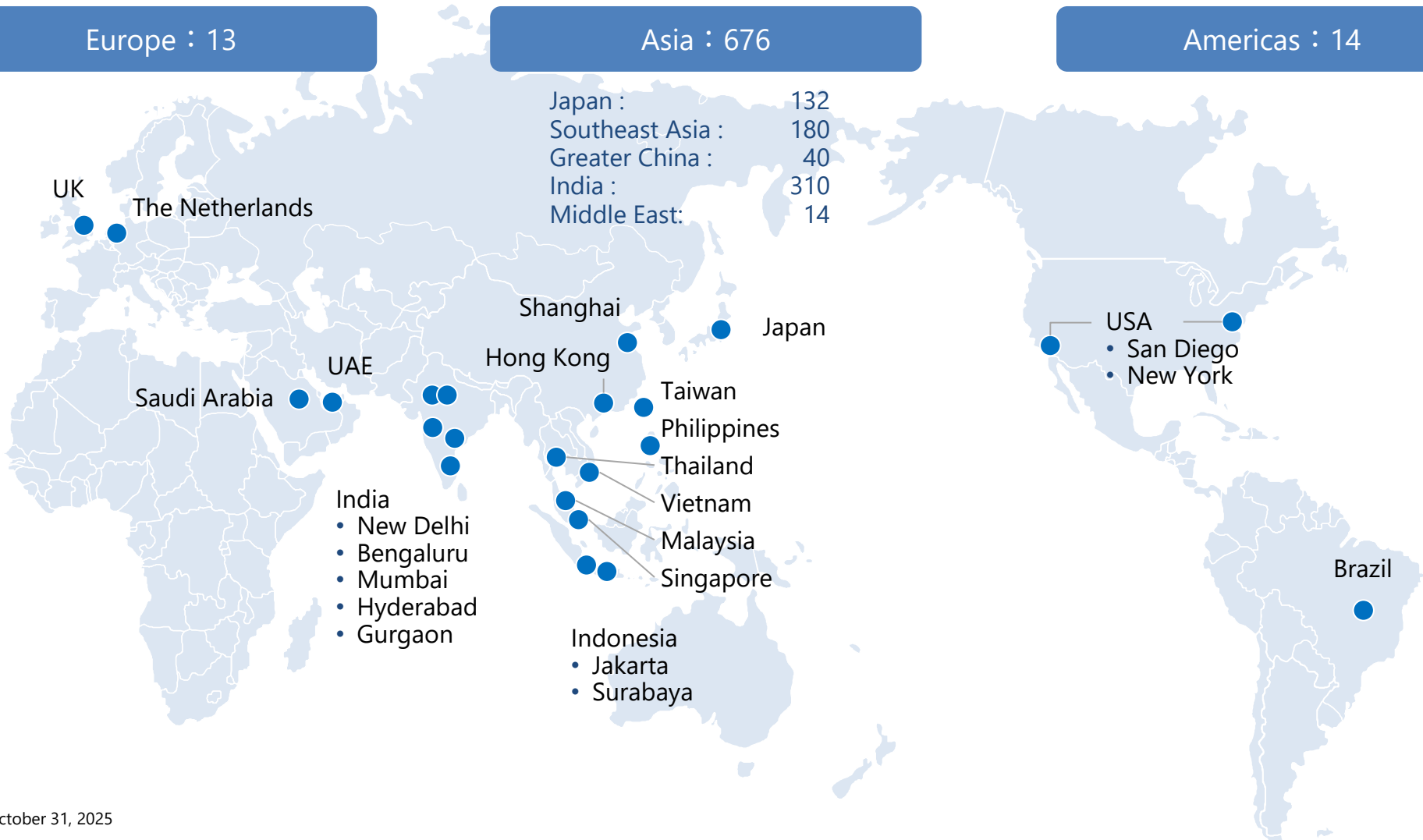
24
Offices

10,000+
Projects Executed

Europe : 13

Asia : 676

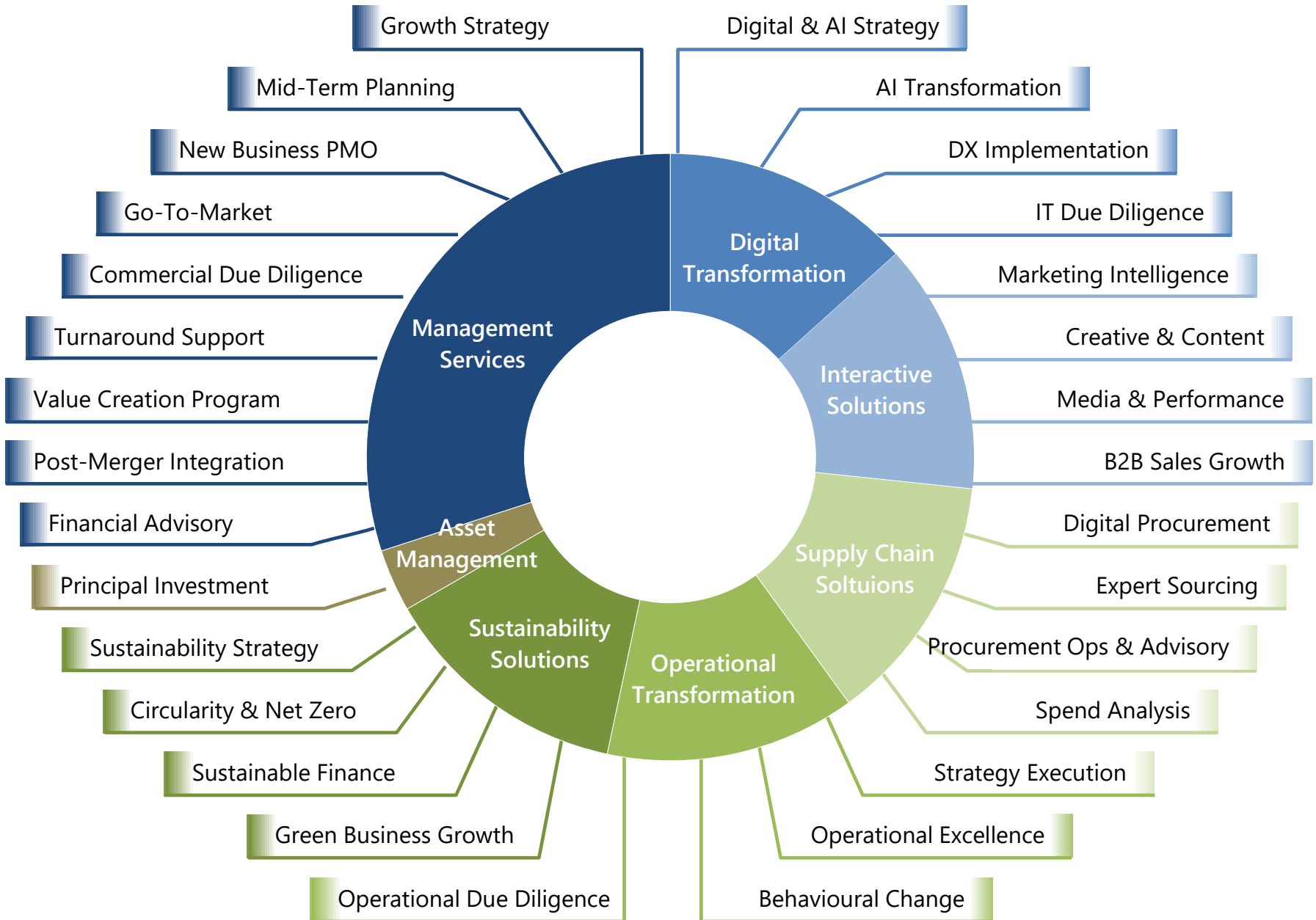
Americas : 14



*As of October 31, 2025

Service Offerings

- We leverage our global network of seasoned professionals, who are equipped with necessary industry know-how and expertise to drive end-to-end transformation for our clients



Our Approach

- We offer deep local expertise and a strong focus on implementation services tailored to unique needs.
- While the industry relies on standardized frameworks and theoretical models, YCP delivers actionable solutions that drive real transformation. Our lean operations ensure personalized attention and efficient management, maximizing your investment's value.

Execution-focused

We balance the possible with the probable – *derive achievable solutions and strategies considering the business fundamentals.*



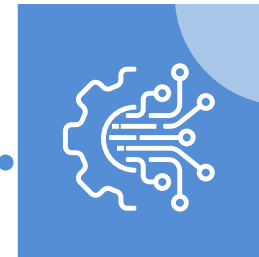
Global Presence

We bring to focus what matters globally – *derive major success factors & pain points of the global strategy and how we can navigate through them across regions.*



Accelerated Innovation

We embed a culture of transformation – *enhance productivity by empowering people, leveraging technology and digital tools.*



Partnering Mindset

We enable collaborative effort – *provide continued support right from strategy inception to execution and create long-term relationships.*



YCP : your Strategy Delivered.

Industry Coverage & Clientele

- 3,000+ project experience across Asia, with Fortune 500 and Asian Conglomerates

Automotive & Mobility



Construction



Energy & Materials



Industrials



TMT



Investment & Finance



Logistics



Infrastructure



Consumer & Retail



F&B



Healthcare

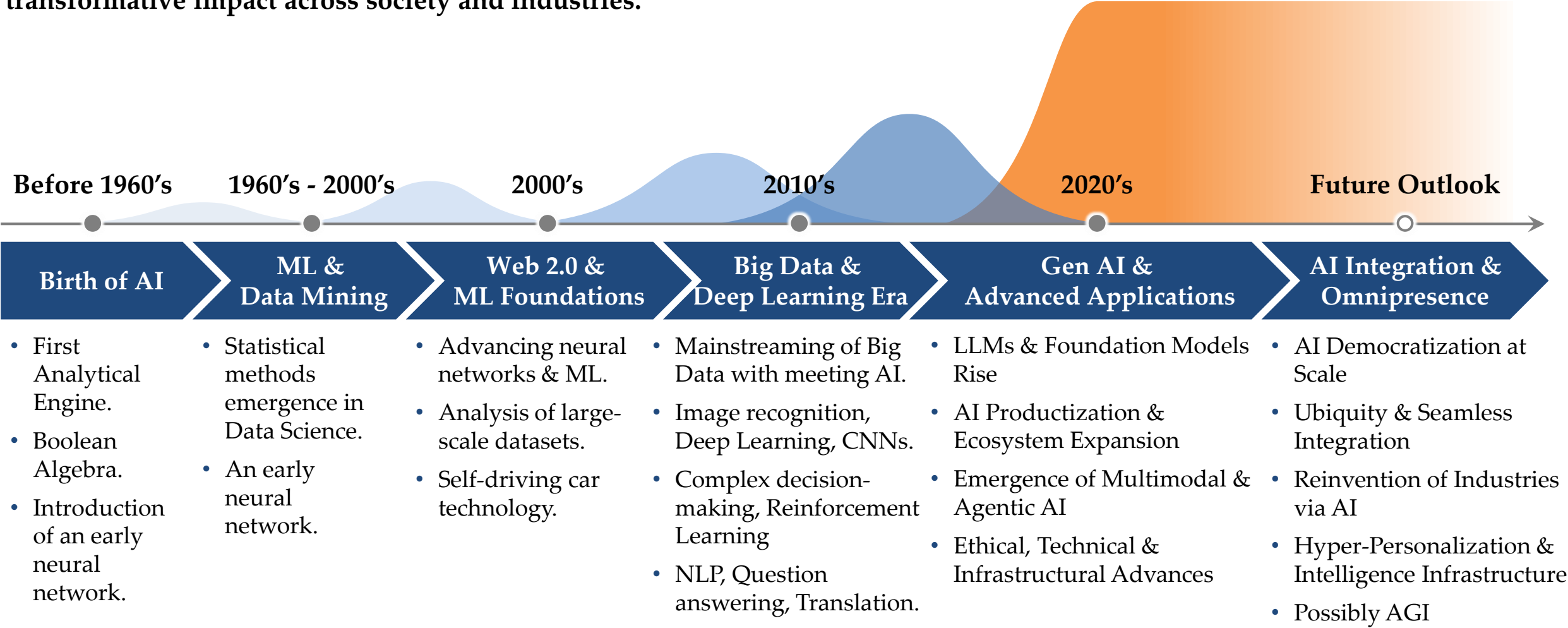


Public Sector



Evolution of AI & Data Science (1/3)

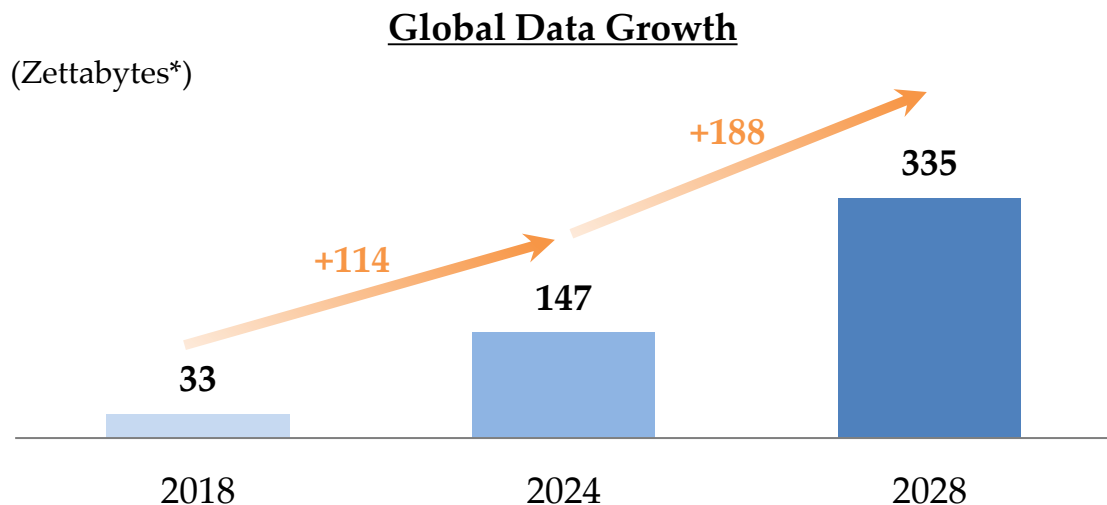
AI's rapid evolution accelerated in the 2020s, unlocking generative intelligence and setting the stage for widespread, transformative impact across society and industries.



Evolution of AI & Data Science (2/3)

AI & Data Science are the fundamental technologies & catalysts for the next industrial revolution

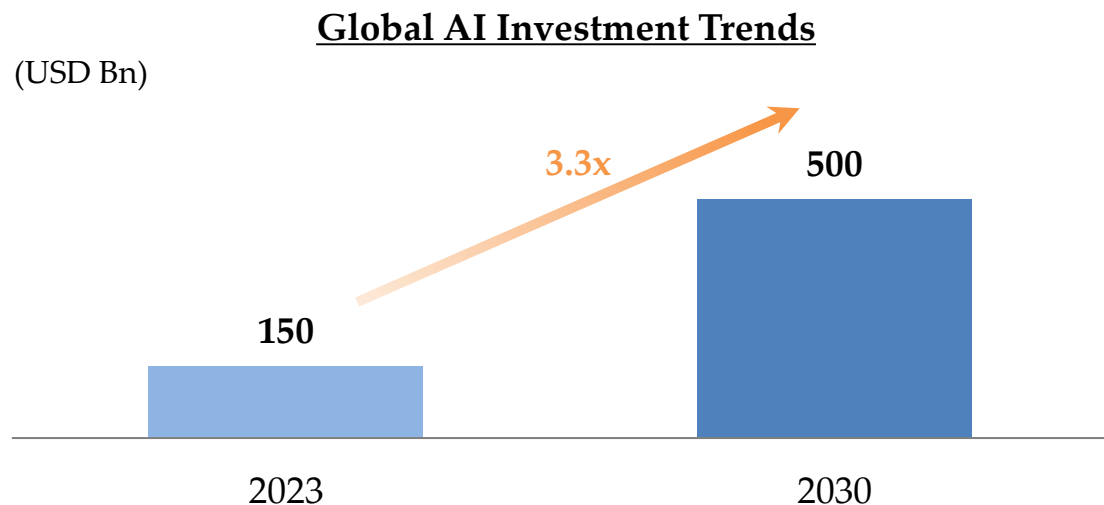
Data: The New Corporate Asset



- **60%** of the world's data created post **AI revolution** in last 2 years
- World's online data to double in less than 4 years.
- By 2026, **75%** of enterprises are projected to adopt AI-powered data governance systems.
- Walmart processes **>2.5 petabytes** of data per/ hr from customer transactions.

* 1 zettabyte (ZB) = 1 trillion gigabytes (GB).

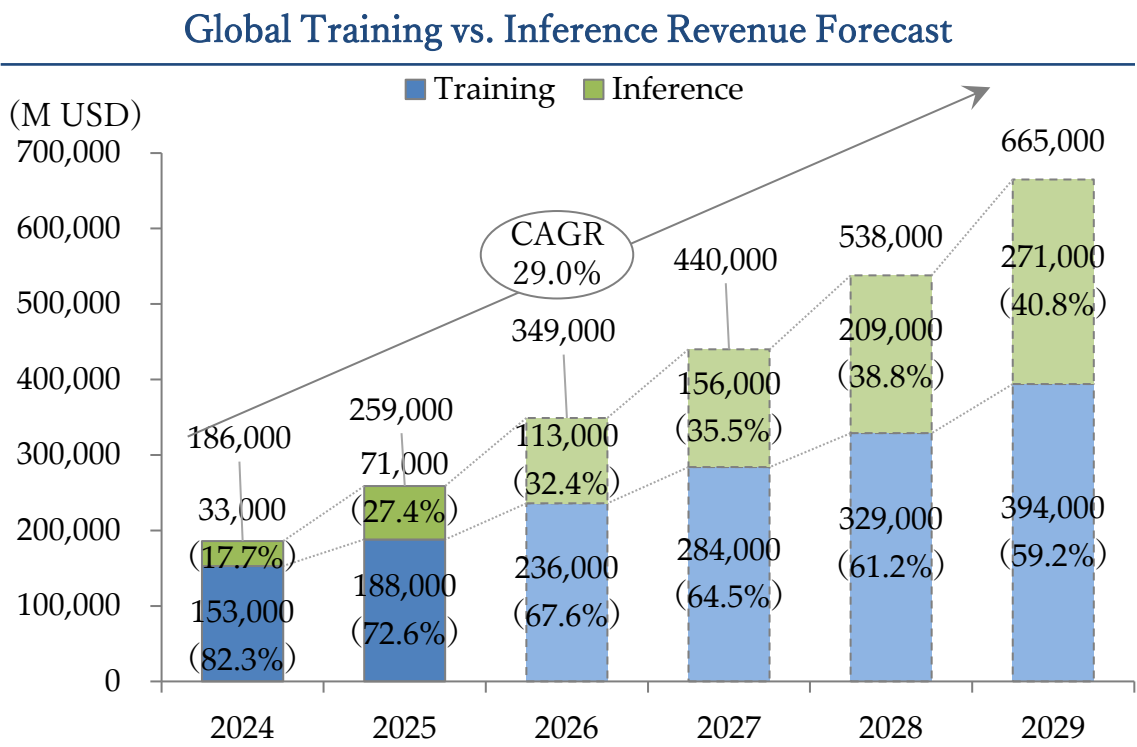
AI: The New electricity?



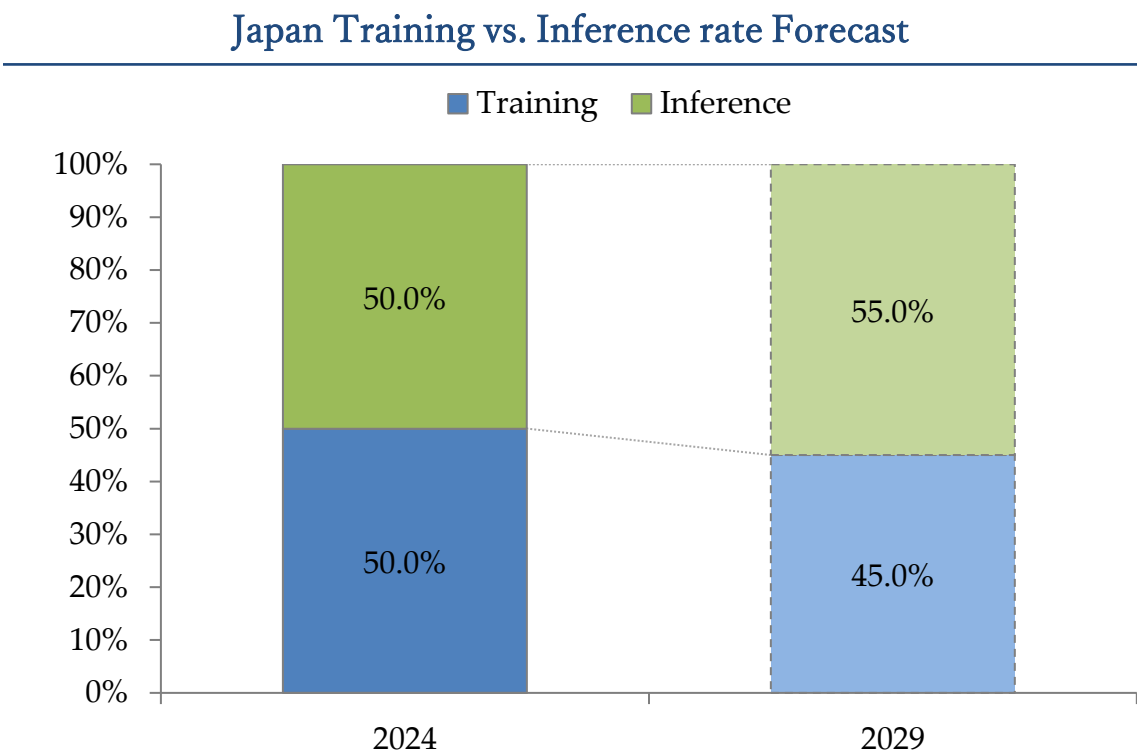
- The number of AI startups grew by **14%** annually from **2020 to 2023**.
- AI startups raised **\$42.5 bn** in 2,500 equity rounds in 2023 alone.
- Companies are deploying AI at the edge (e.g., IoT devices) to reduce latency and enhance real-time decision-making.
- **AI-Powered Automation:** By 2025, **50%** of repetitive tasks in business processes will be automated using AI.

Evolution of AI & Data Science (3/3)

Against the backdrop of heightened interest in generative AI, learning demand currently accounts for the majority. As generative AI matures, it is anticipated that demand will shift towards inference both globally and domestically.



- Whilst demand for training on large-scale language models continues to grow, demand for inference will surge in the future, increasing its share of the overall total.



- The utilization amount is projected to triple over five years (absolute figures undisclosed). Going forward, demand for inference is expected to experience high growth alongside the development and utilization of industry/business-specific models and the maturation of generative AI.

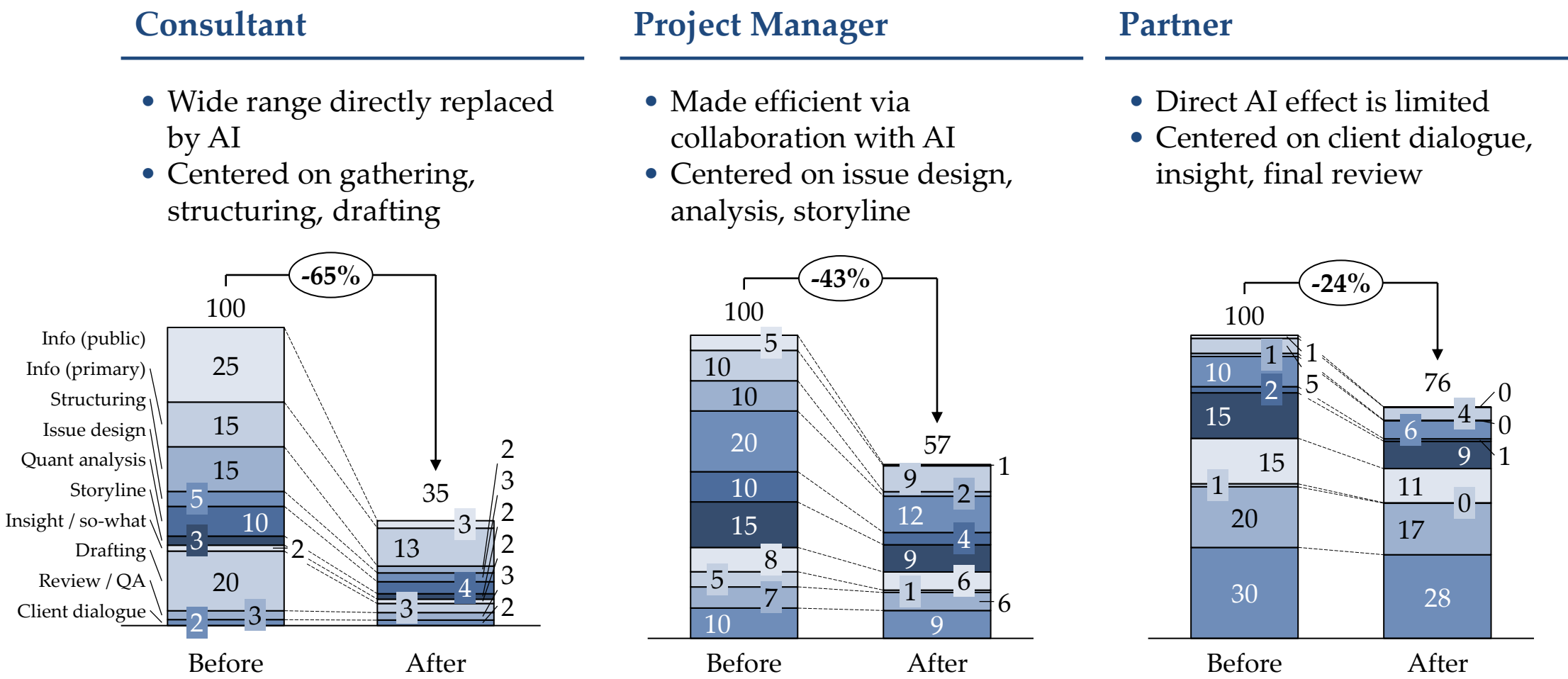
Productivity Gains or Threats from AI (1/2)

Not every human task can be automated. The ways AI and human work fall into five categories as below. It is not a binary of “dump it on AI” vs. “humans carry it all”; we need to choose the best pattern for each task.

Category	What it means	Typical example	Possible Replacement by AI
I. AI alone	AI completes it; humans only review.	Web research, data collection, first-pass analysis	High
II. AI → human	AI designs/prepares; a human executes.	AI drafts interview guide → human runs it	↑
III. Collaboration	AI and human iterate in real time, deepening together.	Issue design, storyline discussion	
IV. Human → AI	A human gathers data; AI analyzes/structures.	Pull paid database → AI analysis	↓
V. Human alone	Only a human can do it; AI cannot.	Client negotiation, final judgment	Low

Productivity Gains or Threats from AI (2/2)

This is a case study at consulting industry. The more junior the role is, the larger share AI makes more efficient. And across all roles, research and analysis shrink, while thinking, coaching, and facing the client increase.



How AI & Data Science are Impacting the Industries

AI & Data Science are driving business transformation across industries, particularly around Manufacturing and Retail sectors, boosting efficiency, accuracy and outcomes while delivering tangible impacts.



Healthcare

- Predictive analytics for disease prevention, personalized medicine, & drug discovery.
- **IBM Watson Health** assists in cancer diagnosis & treatment recommendations.
- **Impact:** Reduced diagnosis time and improved patient outcomes.



Manufacturing

- Predictive maintenance, quality control, and supply chain optimization.
- Siemens uses AI to predict equipment failure and improve production efficiency.
- **Impact:** Decreased downtime and cost savings of **15-20%**.



Agriculture

- Precision farming, crop yield prediction, and pest management.
- John Deere uses AI-powered sensors for monitoring soil conditions and crop health.
- **Impact:** Increased crop yield and sustainable resource usage.



Retail

- Dynamic pricing, inventory optimization, and personalized marketing.
- Amazon uses AI for recommendation engines and supply chain optimization.
- **Impact:** Enhanced customer satisfaction and increased sales by **20-30%**.



Finance

- Fraud detection, credit risk assessment, and algorithmic trading.
- JP Morgan Chase uses AI to detect fraudulent transactions in real time.
- **Impact:** Enhanced security and reduced financial losses.



Transportation

- Autonomous vehicles, route optimization, and demand forecasting.
- Tesla leverages AI for self-driving cars and route planning.
- **Impact:** Improved safety and operational efficiency.

1 Process Automation - E-commerce Platform

Challenges for Client

- Rapid scale-up required closer tracking of developer output and code quality
- Manual-heavy tasks in HR, Finance, PMO, and Marketing needed automation
- Scattered metrics and siloed data led to inconsistent KPI reporting

Impact

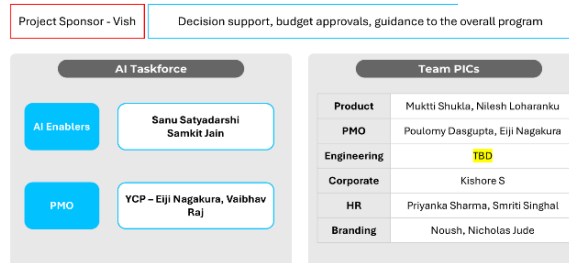
Tech-enabled Approach



AI Taskforce & Mapped Workflows:

- Formed AI Taskforce and mapped tasks across teams.

Team Structure and Ownership



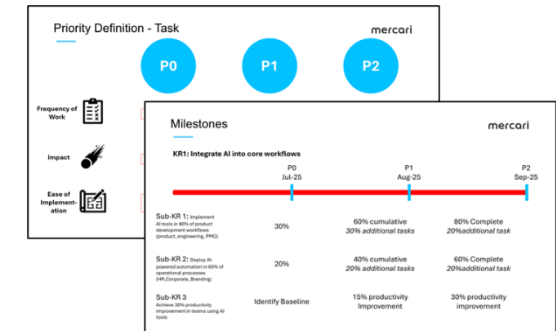
Cross-Functional AI Workshops

- Ran cross-functional AI brainstorming and validation workshops



Data Analysis & Validation

- Built phased rollout roadmap with clear governance and KPIs



- Identified 100+ tasks that can be Automated by AI, leading to time saved and strategic impact.

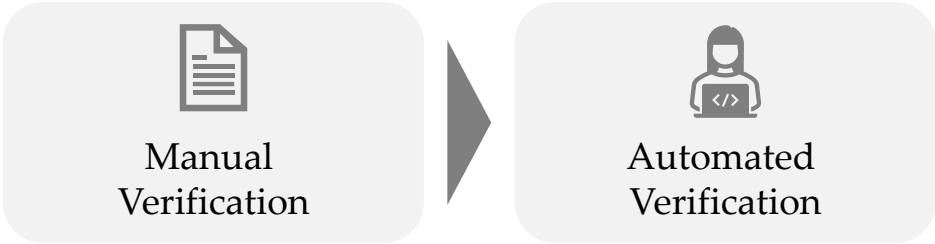
- 25+ automation ideas; high-ROI initiatives for fast-track rollout

- Launched 10+ initiatives; early wins driving **30% productivity boost and process efficiency**

2 Process Automation - Public Sector

Challenges for Client	• E-Governance department of an Indian state. • Need to manually verify millions of documents – citizen applications against Identity and other KYC supporting documents – every month			
Tech-enabled Approach	• Implemented computer vision & NLP models to extract information from supporting documents. • Developed a prototype to identify document types and extract relevant fields for easy access by officers. • The next step involved evaluating the prototype's performance and planning for a full-scale implementation of the solution.			
Process		Headcounts	Time Spent / Day	Annual Cost
	Manual Process	25-30	4-6 hrs	\$ 275,000
	AI-led Process	10-15	1-2 hrs	\$ 65,000

Solution Implementation



Technologies used:

- Computer Vision: ResNet, TensorFlow, DenseNet
- LLM (Open AI GPT 4o)
- OCR: PaddleOCR

Reduced operational costs by ~75%.

- Achieved a **65%** reduction in **TAT** for service applications, significantly improving service delivery efficiency.
- Streamlined workforce need with a **40%** reduction in employees

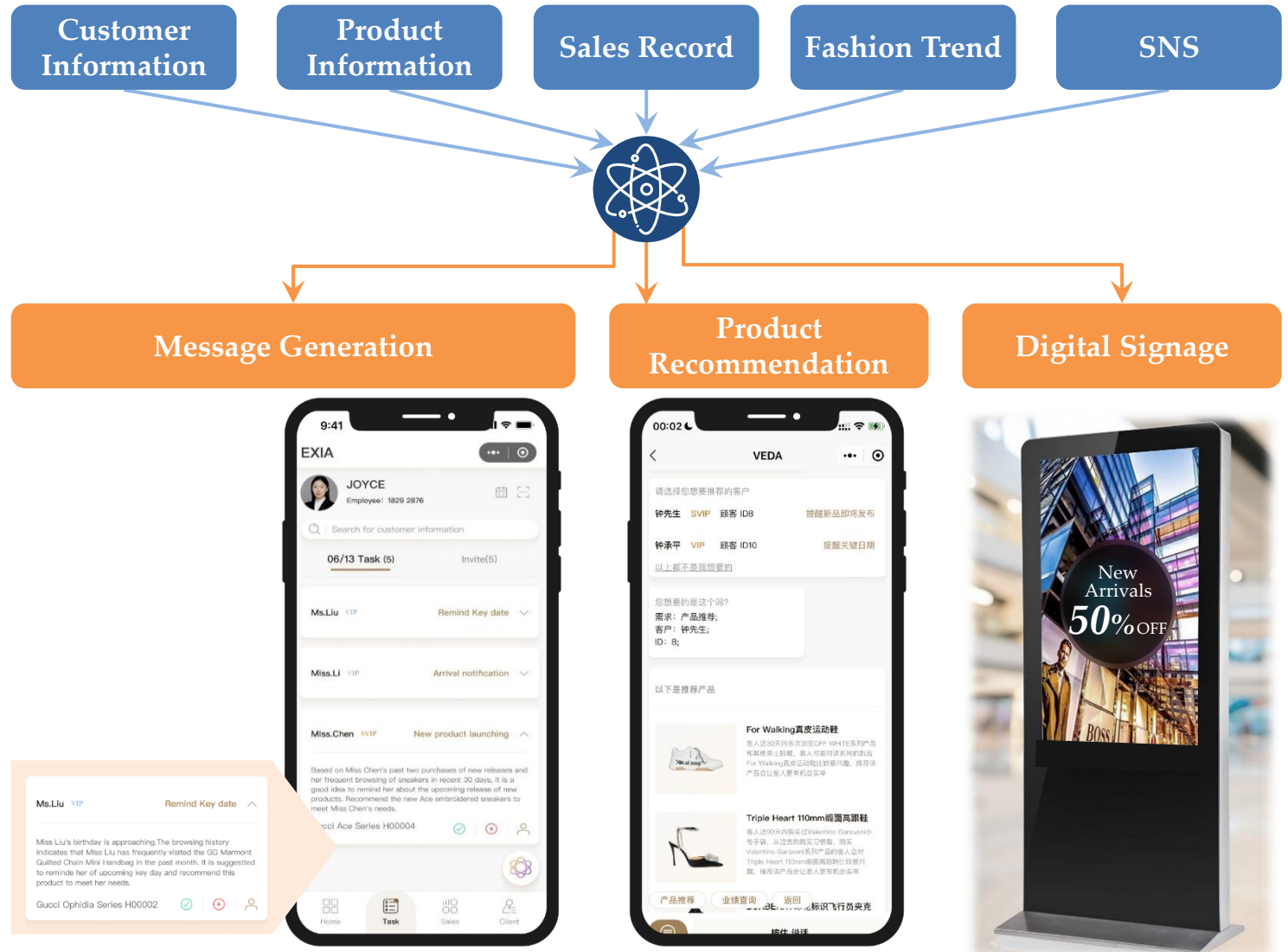
3 Smart CRM to Improve Customer Spend – High-Brand Retail

Challenges for Client

- Most customer data is owned by customer advisor (“CA”)
- No effective tools for CA, training vs real-time experiences
- Difficulties to standardize CA experiences and capabilities
- Focus based on CA’s personal preferences, not customers’

Tech-enabled Approach

- Smart CA - personal AI assistant to empower CA’s operations
- Key functions include voice record & customer tagging, collection of SNS data, advanced AI decision support and communication generation
- AI generated product recommendation help CAs communicate with their customers with most suitable products with reasons
- CAs receive generated messages and communication plan based on the customer information (such as past purchases and birthday)
- Simultaneously, generates marketing contents on digital signage, based on the customers in-store behavior



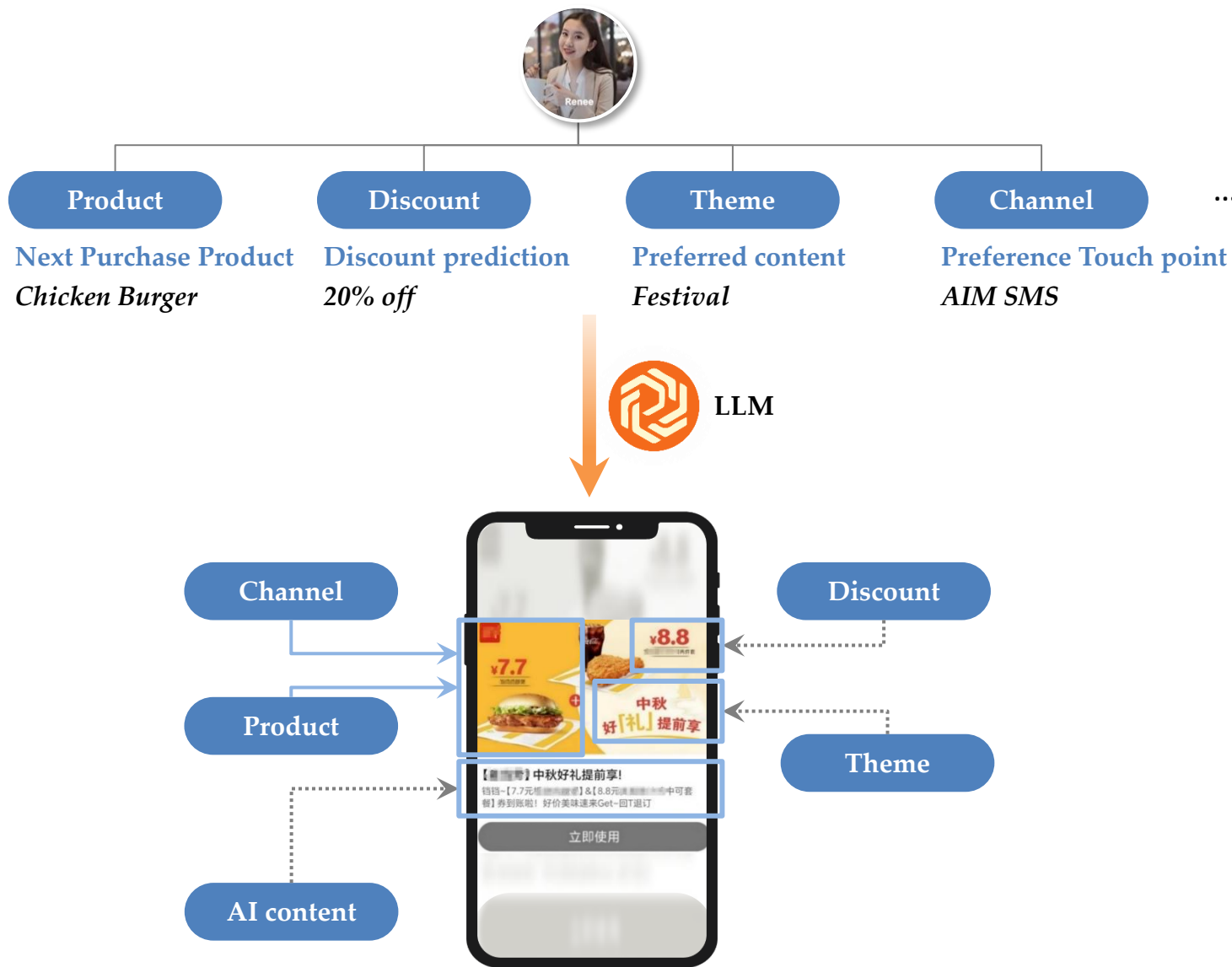
4 Smart CRM for Personalized Communication – Restaurant

Challenges for Client

- Large number of customers with diverse preferences
- Diversified communication scenarios
- Rich and complex product offering

Tech-enabled Approach

- Our AI-powered product provides a seamless & tailored experience for each individual customer
- Key functions include Automated customer life cycle communication & AI generated personalized communication
- Leveraging algorithm to generate personalized communication, incl optimal product/coupon offering, content, timing, and touch points
- 7 days*24 hours automatically operation with industry specific strategy library
- AIGC helps you achieve truly personalized communication with customized content



5 AI Visual Inspection & Automation - Industrials

Challenges for Client

- Severe labor shortages and difficulty in transferring veteran skills (know-how).
- Limitations of conventional rule-based systems in overcoming environmental changes (lighting, temperature, etc.) and complex defect variations.
- Operations tend to be "person-dependent" due to complex natures requiring specific expertise.

Tech-enabled Approach

- Leveraging Deep Learning to achieve human-level precision as well as robustness against environmental noise.
- We provide one-stop solution covering everything from optical device selection to AI integration into production lines.
- User-friendly UI, enabling on-site staffs to build and maintain models without coding.

AI Solution

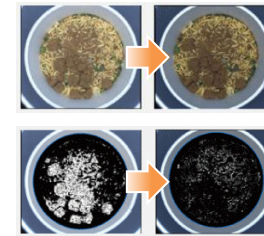
Traditional Rule-Based System

- Issue: Highly Sensitive to Environment
 - Changes in lighting or position can cause errors.
 - Cannot distinguish between harmless dirt and actual defects.

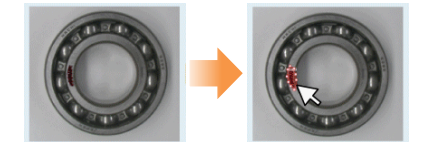
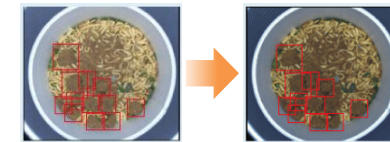
AI-Driven Approach

- Solution: Highly Robust & Accurate
 - Robustness: detects only the target defects (e.g., scratches) even under varying conditions like stains.
 - Accuracy: handles individual product differences effectively.

Robustness



Accuracy



Democratization of Inspection

The system enables anyone to create high-precision models simply by drawing boxes around defects. This "No-Code" approach eliminates the need for specialized skills and prevents person-dependency.

Advanced Automation

Utilizing optimal AI algorithms (Classification, Detection, Segmentation, Anomaly Detection) to automate complex visual inspections that previously relied on human senses.

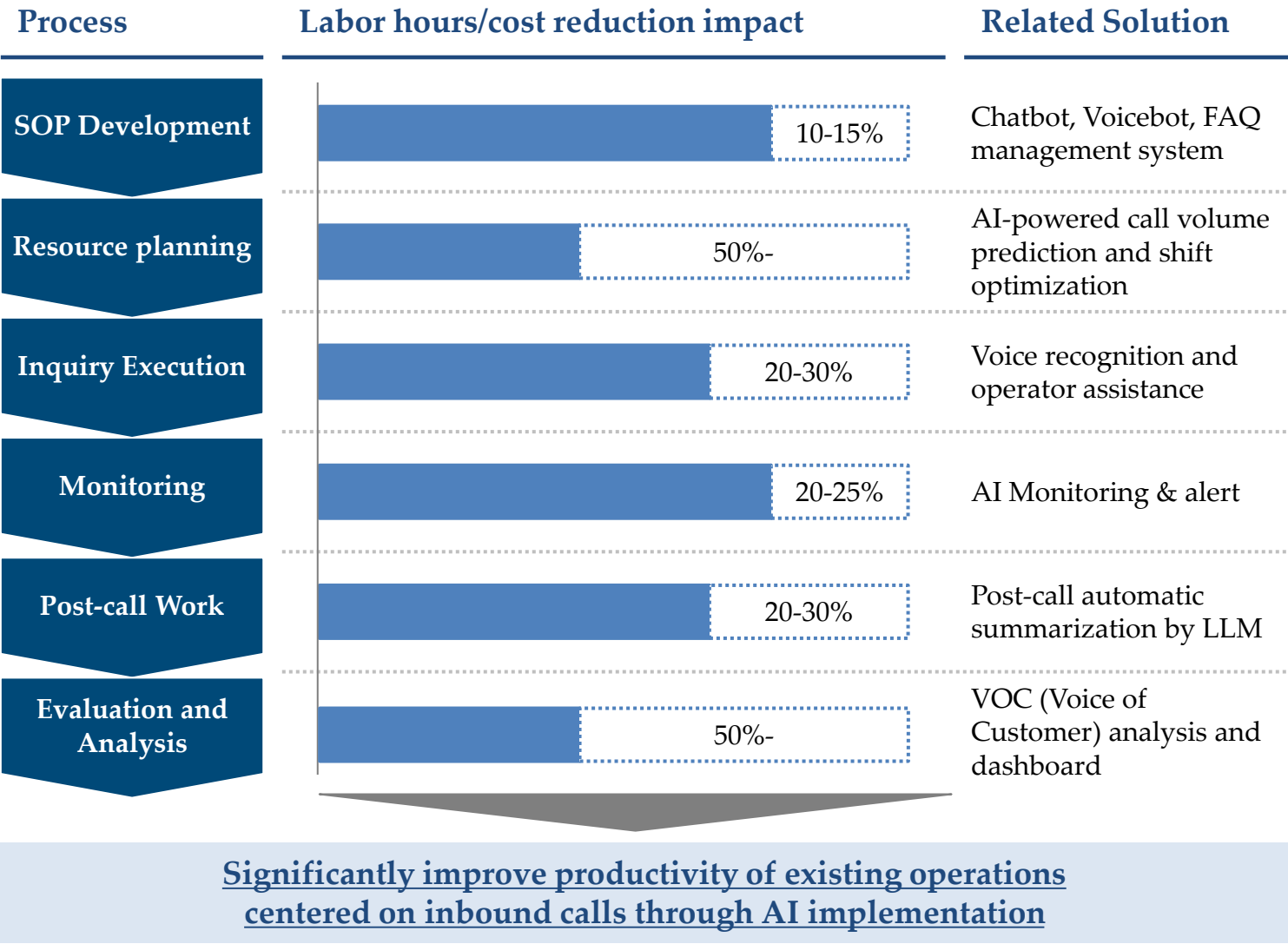
6 AI-Powered Contact Center - Insurance

Challenges for Client

- Insurance products are complicated, so customer questions are difficult and time-consuming to answer
- It is hard for all agents to have the same level of deep knowledge, leading to different qualities of service
- It is difficult to keep trained and experienced agents, which increases costs for hiring and training new staff.

Tech-enabled Approach

- Combined use of Voicebot, Chatbot, and FAQ management system
- AI-powered call volume prediction, shift optimization and operator assistance
- Post-call automatic summarization and automatic category classification
- Post-call conversation analysis and service quality evaluation
- VOC (Voice of Customer) analysis and dashboard



7 AI Interview Platform – Consumer Goods

Current Challenges

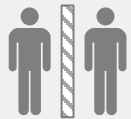


Limitations of Human-led Interviews



Resource Constraints

Time wasted on scheduling, travel, and meeting minutes.



Psychological Barriers

Nervousness in face-to-face settings; honest opinions are withheld.



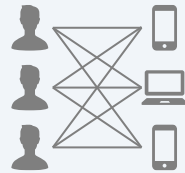
Information Black Box

Knowledge and insights remain stuck in individuals' heads.

Solution

Automated "Dry Interviews" (Factual & Quantitative Data)

1. Input: Scalable Collection



Method:

Asynchronous Interviews

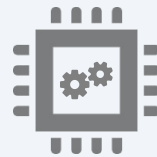
Features:

Utilize spare time; pause and resume freely

Design & Execution:

AI-automated design, interactive deep-dive questioning

2. AI Processing



- Summarization & Generation (NLP)
- Clustering (Trend Analysis)
- Insight Extraction (Issue Discovery)

3. Decision-Support Output



QA Table
(Raw Voices)



Network Diagram
(Correlation Visualization)



Project View
(Overall Trends)



AI Chat , Report
(Deep Dive Analysis)

The system automates the entire interview works



Humans focus on true "Wet Interviews" (Trust, Emotion, Context)

Value Proposition

The efficiency and quality of interviews have dramatically improved, contributing to decision-making.

Productivity:

10 times improvement

Missed Requirements:

75% Reduction

Research Duration:

90% Reduction



Assetization of Tacit Knowledge

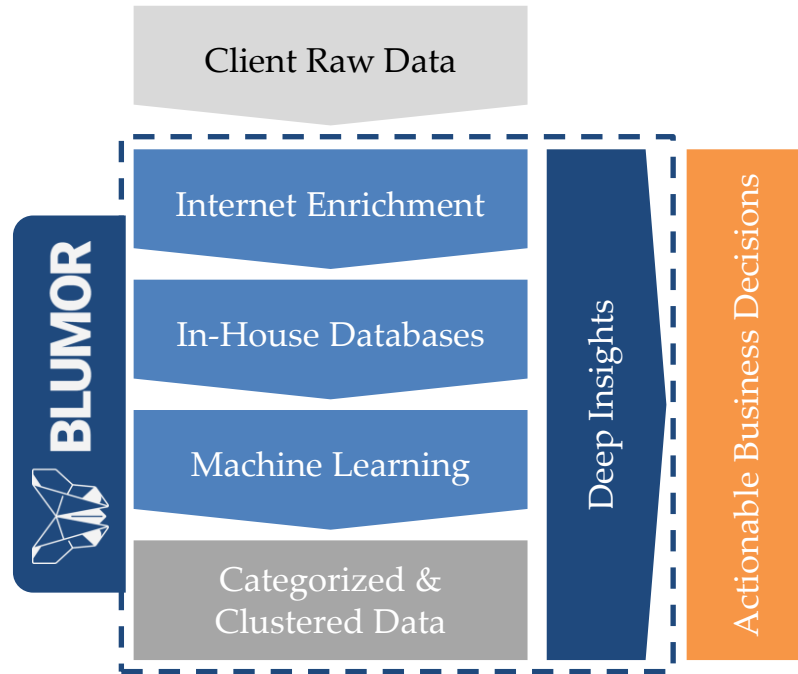
Company-wide sharing of veteran expertise and on-site issues.



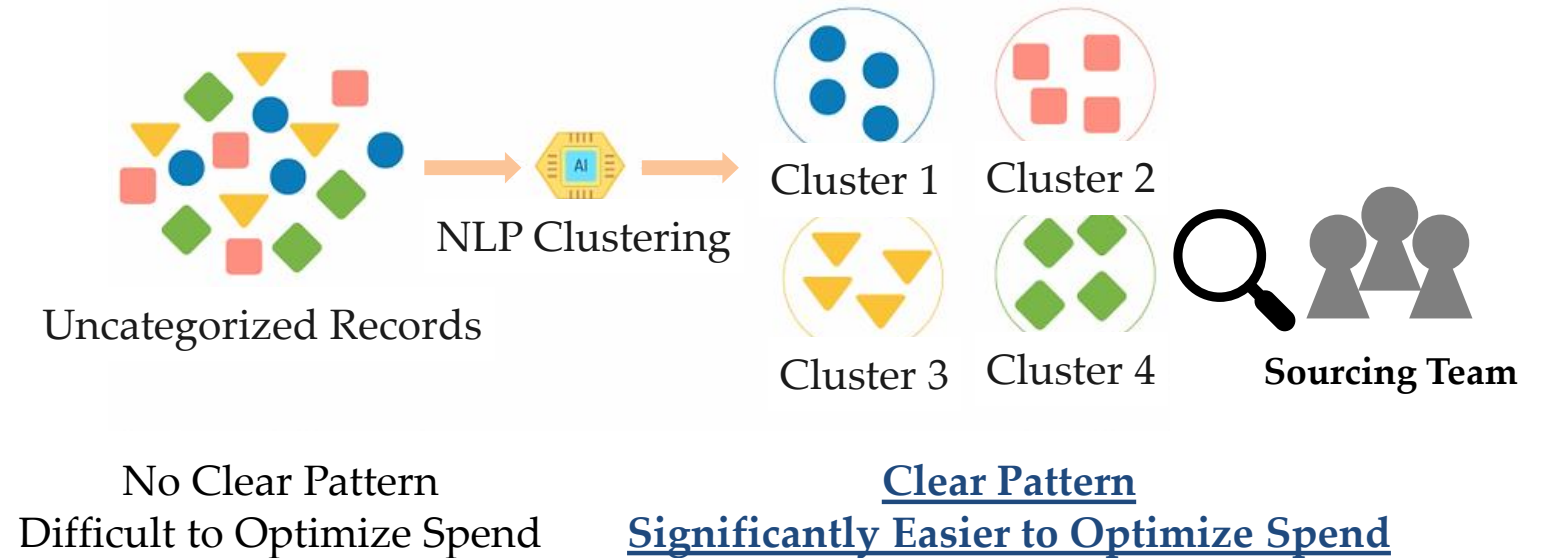
Improving Quality of Decisions

High-precision judgment based on comprehensive data.

8 Blumor – YCP's Proprietary AI algorithm



Clustering



Case Study

NA Waste Management

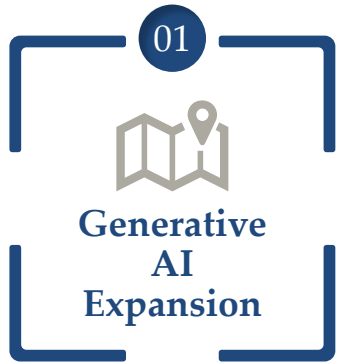


- SKU Categorization
- Inventory Grouping of Suppliers, Locations and Categories

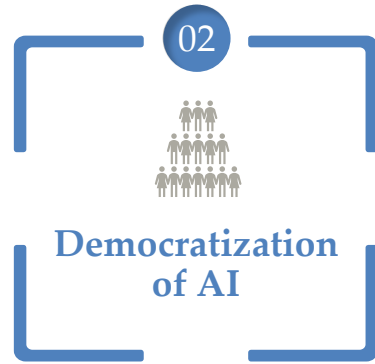


- Categorized 100% spend
- Consolidated product in 30 clusters representing 85% spend leading to: demand consolidation, vendor rationalization & reduction in price and terms variance
- Realized cost savings for client.

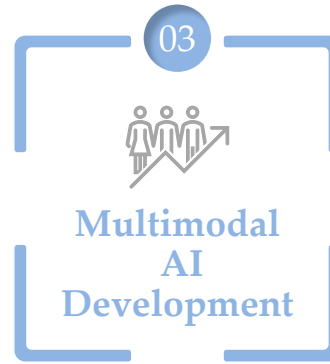
Future of AI and Data Science: Emerging Technologies & Trends



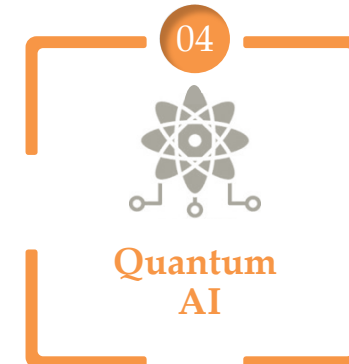
- **Automation:** Streamlining workflows & enhancing productivity.
- **Creative Outputs:** AI-driven content creation.
- **Decision-Making:** AI tools for optimization.



- **Low/No-Code AI:** Tools accessible to non-experts & citizen developers.
- **Wider Access:** Affordable, cloud-based AI all.
- **Open Models:** Publicly available LLMs and datasets.



- **Text, Images, Audio:** Processes diverse data types simultaneously.
- **Cross-Domain AI:** Integrates from multiple modalities.
- **Advanced Analysis:** Enables smarter, decision-making.



- **Quantum Computing:** Tackles problems classical computers can't solve.
- **Quantum ML:** Combines quantum power with AI models.
- **Large-Scale Chips:** Advances quantum hardware scalability.

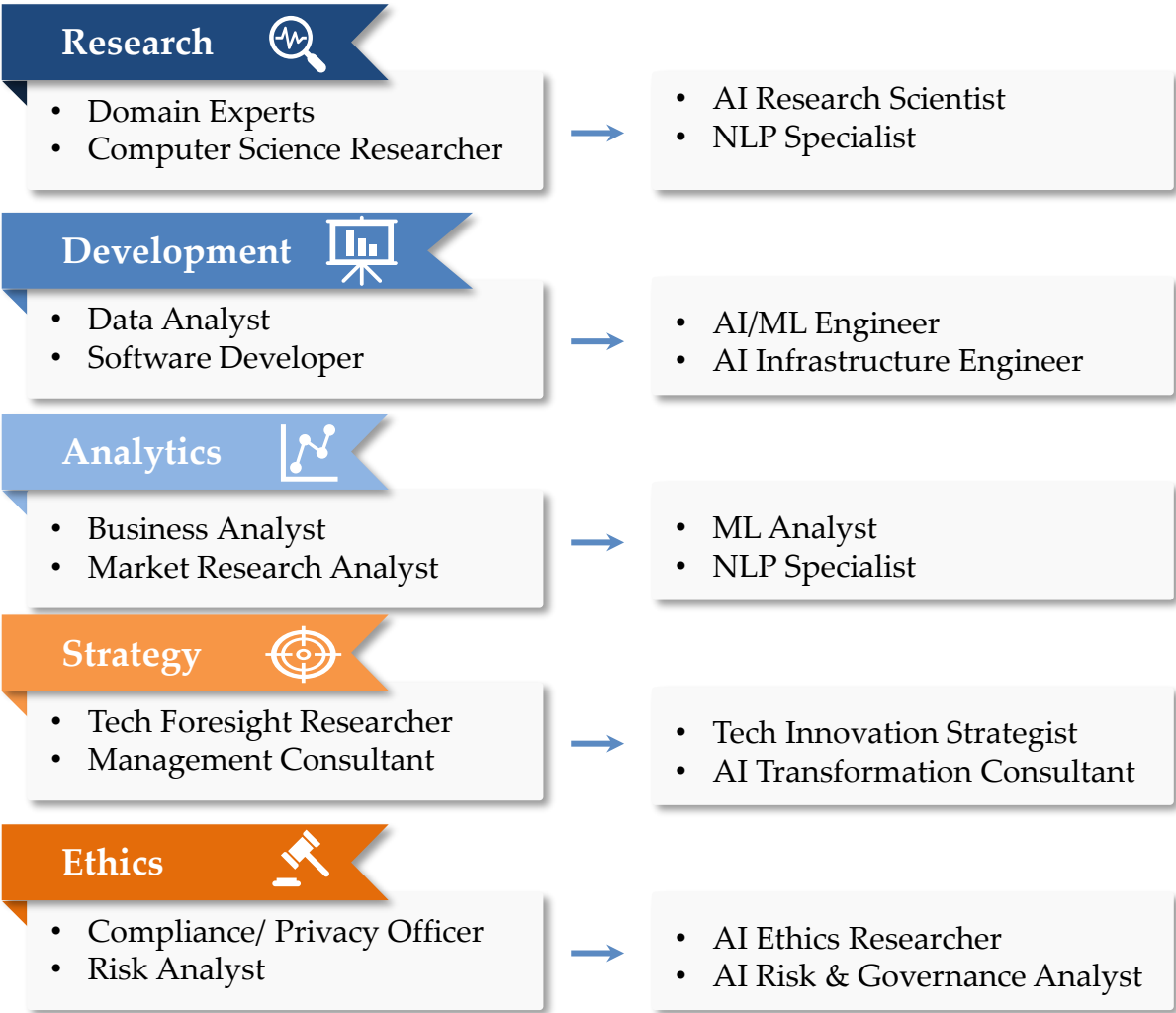


- **Ethical AI:** Ensures fairness & inclusivity in AI solutions.
- **Transparency:** Builds clear, understandable AI systems.
- **Regulatory Compliance:** Adheres to global AI standards.

Emerging technologies in AI & Data Science are **driving democratization automation, advanced analytics, and ethical AI**, while integrating quantum computing & multimodal capabilities to revolutionize industries and enhance decision-making.

Transforming Career Landscapes and Demands

Change in Traditional Career Paths



Impact on Career Opportunities

- **Emerging Job Roles:** AI Specialist, Data Scientist, Machine Learning Engineer, AI Product Manager, and Ethical AI Officer.
- **Industry Growth:** AI and Data Science job postings grew by **74% annually in the past four years.**
- AI talent demand is highest in sectors like **Finance, Healthcare, Retail, and Autonomous Systems.**
- Data Science and AI professionals **earn 1.5 to 2.5 times** more than traditional IT roles.

Skill Requirements

- **Technical skills** like Python, R, SQL, and ML frameworks remain foundational for AI roles, alongside data, solid grasp of linear algebra, probability, and statistics.
- **However, the rise of no-code and low-code AI tools is shifting the skill landscape.** Soft skills like Communication, creativity, and product thinking are emerging as critical differentiators.
- **New-age professionals are required to understand AI ethics, data privacy laws, and model transparency..**

